

COURSE TITLE : MAN MADE FIBERS
COURSE CODE : 6063
COURSE CATEGORY : A
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

Module	Topics	Periods
1	Introduction to Man- made Fibers	19
2	Polyamides & Polyesters	19
3	Recently Developed manmade Fibers	19
4	Blending , Texturisation and Processing	18
TOTAL		75

COURSE OUTCOME

Module	Go	Student will be able to
1	1	Understand the classification of manmade fibers & different types of polymerizations
	2	Know the different methods of spinning and the drawing process.
2	1	Understand the manufacture, properties & uses of nylon 66 & nylon 6
	2	Know the manufacture, properties & uses of polyester
3	1	Understand the manufacture, properties & uses of polyolefin fibers .
	2	Know the manufacture, properties & uses of acrylic, PVA, Polyurethane fibers & Evaluate the properties of Nomex and Kevlar fibres.
4	1	Know the Blending natural and synthetic fibres.
	2	Understand the different process of crimping of manmade yarns.
	3	Know the changes required for processing man-made fiber in the spinning & weaving machineries & size recipe

SPECIFIC OUTCOME

MODULE I INTRODUCTION TO MAN- MADE FIBERS

- 1.1.0 Understand the classification of manmade fibers & different types of polymerizations**
- 1.1.1 Classify the important fully synthetic fibres
 - 1.1.2 Define the terms monomers, polymers, polymerization and degree of polymerization.
 - 1.1.3 List the classification of Polymers.
 - 1.1.4 Describe condensation, addition and co - polymerization.
 - 1.1.5 Understand the random, alternate, block co-polymers

1.2.0 Know the different methods of spinning and the drawing process

- 1.2.1 Explain the different methods of spinning man made filaments.
- 1.2.2 Describe the drawing process and properties accrued by drawing the filaments.
- 1.2.3 Illustrate the working of a draw twist machine.
- 1.2.4 State the importance of staple fibre

MODULE II POLYAMIDES & POLYESTERS

2.1.0 Understand the manufacture, properties & uses of nylon 66 & nylon 6

- 2.1.1 List the different polyamide fibers.
- 2.1.2 Explain the manufacture of Nylon 66
- 2.1.3 List the physical and chemical properties and uses of Nylon 66.
- 2.1.4 Explain the manufacture of Nylon 6.
- 2.1.5 List the properties of and uses of Nylon 6.

2.2.0 Know the manufacture, properties & uses of polyester

- 2.2.1 Illustrate the chemical structure of polyester fibres.
- 2.2.2 Explain the manufacture of Polyester filament Yarn and staple fibre.
- 2.2.3 Describe the important physical and chemical properties of polyester and its uses

MODULE III RECENTLY DEVELOPED MAN MADE FIBERS

3.1.0 Understand the manufacture, properties & uses of polyolefin fibers

- 3.1.1 List recently developed synthetic fibres.
- 3.1.2 Describe the method of production of LDPE & HDPE.
- 3.1.3 Differentiate between LDPE and HDPE.
- 3.1.4 State Properties, uses and microscopic appearances of LDPE and HDPE.
- 3.1.5 Describe the methods of manufacture of Polypropylene.
- 3.1.6 State the properties, uses and structure of Polypropylene

3.2.0 Know the manufacture, properties & uses of acrylic, PVA, Polyurethane fibers & the properties of Nomex and Kevlar fibres

- 3.2.1 Distinguish between Acrylic and Modacrylic fibres.
- 3.2.2 Describe the method of production of Orlon fibres.
- 3.2.3 State the structure, properties and uses of Orlon.
- 3.2.4 Explain the method of manufacture of Polyurethane fibre.
- 3.2.5 State the structure, properties, and uses of Spandex fibre.
- 3.2.6 Explain the method of manufacture of PVA, properties and uses.
- 3.2.7 Evaluate the properties of Nomex and Kevlar fibres

MODULE IV BLENDING , TEXTURISATION AND PROCESSING

4.1.0 Know the Blending natural and synthetic fibres

- 4.1.1 State the importance of blending natural and synthetic fibres.

- 4.1.2 List the general characteristics of blended material
- 4.1.3 Explain the two different methods of blending
- 4.2.0 Understand the different process of crimping of manmade yarns**
 - 4.2.1 Define texturisation and its objectives
 - 4.2.2 Differentiate stretch yarn, modified stretch yarn and bulk yarn
 - 4.2.3 Describe the principle and working of a false twist texturising machine
 - 4.2.4 Explain the working of Stuffer Box crimping machine
 - 4.2.5 State the working of Edge crimping machine.
 - 4.2.6 Explain the Air – Texturisation process and its salient features
 - 4.2.7 Describe knit – de – knit texturising process.
 - 4.2.8 Explain the working of gear crimping machine
- 4.3.0 Know the changes required for processing man-made fiber in the spinning & weaving machineries & size recipe**
 - 4.3.1 List out the necessary changes required for processing man-made fiber in the existing spinning line.
 - 4.3.2 List out the changes required for processing man-made fiber in the existing weaving shed
 - 4.3.3 List the size recipe for sizing man – made fibre and their blends
 - 4.3.4 Control the static, temperature and humidity condition while processing man-made fibers

CONTENT DETAILS

MODULE I INTRODUCTION TO MAN – MADE FIBRES

Different classification of fully synthetic fibers- Polyamide (Nylon 66, Nylon 6)- Polyester (Terylene, Dacron)- Poly olefin (Poly ethylene and polypropylene)- Poly Acrylonitrile (Acrylic, Mod Acrylic)- Poly vinyl alcohol- Poly Urethane fiber (Spandex)- Aramid fibers (Nomex & Kevlar)- Definition of monomers, polymers, polymerization, Degree of polymerization-Classification of Polymerization- Explain Condensation polymerization, Addition polymerization and Co-polymerization – their examples with equations- Random, Alternative, Block co-polymers- Linear, branched, and cross linked polymers. Principles of spinning and methods of- Melt spinning, Wet spinning, Dry spinning. Sketch and explain each method - Advantages and disadvantages and suitability. Importance of Drawing or Stretching- Advantages- Draw ratio- Filament ratio- stretch ratio cold drawing- hot drawing- Explain Draw twist machine with sketch- Importance of Staple fiber.

MODULE II POLYAMIDES AND POLYESTERS

Chemistry of Polyamide fiber- Nylon 6 6- Nylon 6- Manufacture of Nylon 6 6-Raw materials- Polymerization Spinning- Drawing- Flow chart with equation- Properties of Nylon 6 6- Microscopic appearance- Physical properties- Chemical properties- Uses of Nylon 6 6. Production of Nylon 6- Production caprolactum- Condensation polymerization- Melt spinning- Stretching- Flow chart with equation- Properties- Microscopic appearance-Physical properties- Chemical properties- Uses.

Introduction of PET fiber & differentiate types- Terylene- Dacron- Manufacture of Terylene-Production of Ethylene glycol and D M T- Polymerization - Line Sketch- Melt spinning- Hot drawing- Flow chart for Filament and staple fiber- Properties- Microscopic appearance-Physical properties- Chemical properties- Uses.

MODULE III RECENTLY DEVELOPED SYNTHETIC FIBRES

Names of recently developed synthetic fiber, Poly ethylene, Poly propylene, Acrylic fibre, PVA Fibre, Poly Urethane Fibre, Nomex Fibre, Kevlar Fiber

Method of manufacturing HDPE- Raw material- Polymerization with Ziegler type catalyst-Melt spinning and Drawing- Properties, uses- Differentiate LDPE& HDPE.

Polypropylene -Three different types of Polymers, Manufacture- Raw material, Polymerization, Melt spinning and drawing- Properties, uses and appearance of Polypropylene.Differentiate Acrylic and Modacrylic- Production of Orlon Fiber- Vinyl cyanide- Continuous polymerization- Wet spinning and dry spinning- Stretching- Properties, uses and appearance of Acrylic fiber.PVA fiber- Manufacture- Raw materials- Polymerization- saponification- Wet and dry spinning- Drawing- Properties and uses of PVA Polyurethane Fiber- Manufacture, properties and uses of Spandex.

Brief study of Aramid, Nomex, Kevlar, Carbon & Glass fibers

MODULE- IV BLENDING AND PROCESSING OF MAN-MADE FIBRES

Blending- Importance- Characteristics of Blended material- Explain the two methods of Blending- Blow room Blending- Draw frame blending- Advantages and disadvantages.

Texturisation- Objectives- Properties of texturised yarn Types of Texturized yarns- Stretch Yarn- Modified Stretch yarn- Bulk yarn- Detailed study of following Texturization methods with sketches- False twist texturization- Stuffer Box crimping- Knife edge crimping- Air texturizing- Knit – De- Knit texturizing- Gear crimping- Knife edge crimping- Detailed study of texturizing machines with the aid of sketches- Advantages and uses of Texturized yarn.

Changes in spinning line while processing man-made fibers- Blow Room- Card- Draw frame-Simplex- Ring Frame. Size recipe for- Polyester filament yarn- Polyester / Viscose- Polyester / cotton, Control of static charges and humidity.

REFERENCE BOOKS

1. R.W.Moneritf Man-Made Fabrics -. Vol. 1, 2, 3 Wood Head Publishing
2. V.A Shenai Textile Fibers - .Vol.1,2
3. Usenko Processing of Manmade fabrics Universal Publishing corp., Mumbai -2
4. -R.R.Chkkaravarthy A Glimpse on the chemical Technology of Textile Fibers
5. R.Jagannathan Polyester Blends and spinning - Vol 4
7. Textile Association Publication Polyester Blend spinning
8. His L. Unsinji P Yarn Texturising Technology Universal Publishing corp., Mumbai -2
9. Wilson D K and Koller T Production of Textures Yarn-